Dr. D. Y. PATIL
College of Engineering and Innovation
APPROVED BY AICTE, RECOGNIZED BY GOVT. OF
MAHARASHTRA & AFFILIATED TO S. P. PUNE UNIVERSITY

A SEMINAR REPORT ON

" Cybersecurity Threat Detection "

Submitted to

Savitribai Phule Pune University, Pune

In partial fulfilment of the requirements for the award of the degree of

Bachelor of Engineering (Computer Engineering)

Submitted By:
Darekar Kalyani Rajendra
Roll No: 13142 (Div – A2)
Exam Seat No:

Under the Guidance of:
Prof. Shirsagar Shubhangi
Department of Computer Engineering
DYPCOEI, Talegaon Varale

Department of Computer Engineering

Dr. D. Y. Patil College of Engineering and Innovation

Varale, Talegaon, Pune - 410507

Savitribai Phule Pune University

Academic Year: 2026-2027 (Sem-V)

Dr. D. Y. PATIL
College of Engineering and Innovation
APPROVED BY AICTE, RECOGNIZED BY GOVT. OF
MAHARASHTRA & AFFILIATED TO S. P. PUNE UNIVERSITY

CERTIFICATE

This is to certify that the seminar report entitled " **Cybersecurity Threat Detection "** submitted by Ms. Kalyani Rajendra Darekar , Third Year Computer Engineering, Roll No: 13142, Div: A2, is a bonafide student of this institute. The seminar work has been carried out under the supervision of Prof. Shirasagar Shubhangi and is approved for partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering (Computer Engineering) of Savitribai Phule Pune University for the academic year 2026-2027.

Prof. Shubhangi Kshirsagar
(Seminar Guide)
DYPCOEI, Varale

Prof. Anita Shinkar
(Seminar Coordinator)
DYPCOEI, Varale

(External Examiner)

Dr. Alpana Adsul
(HOD Computer Engg)
DYPCOEI, Varale

Dr. Suresh Mali
(Principal)
DYPCOEI, Varale

Place: Talegaon, Pune

Date: / /2026

DECLARATION

I declare that this Seminar report entitled **" Cybersecurity Threat Detection "** is my own academic work conducted under the guidance of **Prof. Shirsagar Shubhangi** from the Department of Computer Engineering, Dr. D. Y. Patil College of Engineering and Innovation, Varale, Talegaon, Pune.

To the best of my knowledge and belief, I declare that:

1. The seminar report does not contain any work previously submitted for the award of any degree, diploma, or associateship in this or any other university.

2. The percentage of plagiarism is within the permissible university norms (Max 12–20% as per Savitribai Phule Pune University guidelines).

3. AI-oriented content is verified to be under 10% in accordance with institutional academic integrity mandates.

4. All sources of literature, theoretical frameworks, data standards, and comparative findings have been duly cited and acknowledged via standard IEEE referencing.

Darekar Kalyani Rajendra
Roll No: 13142 (Div – A2)
Student, Department of Computer Engg.

Prof. Shirsagar Shubhangi
(Seminar Guide)
Department of Computer Engg.

ABSTRACT

Cybersecurity threat detection has become a foundational discipline for protecting modern digital infrastructure against an increasingly sophisticated and fast-evolving population of adversaries. Enterprises, cloud platforms, financial institutions, critical infrastructure operators, and government agencies now depend on continuous, automated mechanisms capable of identifying malware, phishing campaigns, ransomware, distributed denial-of-service (DDoS) attacks, and unauthorized intrusions in near real time. Traditional signature-based defences, while effective against previously catalogued threats, are structurally unable to recognise novel or polymorphic attacks, motivating the shift towards anomaly-based, behaviour-based, and machine-learning-driven detection paradigms.

Building a dependable threat-detection capability requires balancing four interdependent pillars:

- Detection Accuracy: minimising both false positives, which overwhelm security analysts, and false negatives, which allow real intrusions to go unnoticed.
- Adaptability: the ability to recognise zero-day exploits and evolving attacker tradecraft without waiting for signature updates.
- Real-Time Performance and Scalability: processing high-velocity network, endpoint, and log data across large, distributed environments without introducing unacceptable latency.
- Explainability and Analyst Trust: presenting detection outcomes in a form that Security Operations Centre (SOC) analysts can verify, prioritise, and act upon.

This report examines the fundamentals of cybersecurity threats, surveys signature-based, anomaly-based, and behaviour-based detection techniques, and analyses the architecture of Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), and Security Information and Event Management (SIEM) platforms. It further explores the growing role of machine learning and artificial intelligence in threat detection, the function of threat intelligence feeds and frameworks such as MITRE ATT&CK, and the practical challenges of deploying detection systems at scale, including alert fatigue, adversarial evasion, and encrypted-traffic visibility. The report concludes with a discussion of future directions, including autonomous response, explainable AI for security operations, and collaborative threat-intelligence sharing, providing a comprehensive technical foundation for understanding and presenting the subject of cybersecurity threat detection.

Keywords: *Cybersecurity, Threat Detection, Intrusion Detection System (IDS), Intrusion Prevention System (IPS), Security Information and Event Management (SIEM), Signature-Based Detection, Anomaly Detection, Behavioural Analytics, Machine Learning, Threat Intelligence, Incident Response.*

ACKNOWLEDGEMENT

I express my deepest gratitude and sincere appreciation to my esteemed seminar guide, Prof. Shirsagar Shubhangi, Department of Computer Engineering, Dr. D. Y. Patil College of Engineering and Innovation, Varale, Talegaon, for invaluable guidance, constructive critique, and constant encouragement throughout the formulation and execution of this seminar report. Her comprehensive technical insights into cybersecurity, intrusion detection, and intelligent security systems substantially broadened my academic perspective and elevated the overall rigour of this work.

I convey my profound gratitude to Dr. Alpana Adsul, Head of the Department of Computer Engineering, and Prof. Anita Shinkar, Seminar Coordinator, for providing exemplary academic infrastructure, laboratory access, and an inspiring institutional environment conducive to advanced technical investigation. I am equally grateful to Dr. Suresh Mali, Principal of DYPCOEI, for his visionary leadership and continuous institutional support.

I also extend my heartfelt thanks to all faculty members, technical laboratory assistants, and library staff of the Computer Engineering Department for their unstinted cooperation and timely assistance in securing critical research literature and standards documentation. My sincere acknowledgment extends to my peers and colleagues for their intellectually stimulating discussions and critical perspectives on modern cybersecurity practice.

Lastly, I owe an eternal debt of gratitude to my parents and family members for their boundless love, patient understanding, moral encouragement, and sacrifices throughout my educational pursuits.

Kalyani Rajendra Darekar

(Roll No: 13142, Div: A2)

Third Year Computer Engineering

TABLE OF CONTENTS

Abstract iii
Acknowledgement v
List of Figures viii
List of Tables ix
Nomenclature & Abbreviations x
1. Introduction 1
1.1 Background and Global Cybersecurity Imperatives 1
1.2 The Evolution of Cybersecurity Threat Detection 1
1.3 The Detection Gap: Alert Volume vs. Analyst Capacity 2
1.4 Motivation and Purpose of the Study 2
1.5 Objectives of the Seminar 3
1.6 Organisation of the Report 3
2. Literature Review 4
2.1 Cybersecurity Threats – Concepts and Taxonomy 4
2.2 Signature-Based Detection in the Literature 5
2.3 Anomaly-Based and Statistical Detection 6
2.4 Behaviour-Based Detection and UEBA 6
2.5 Machine Learning and Deep Learning in Threat Detection 6
2.6 Gap Analysis and Comparative Positioning 7
3. System Architecture and Core Technologies 9
3.1 Signature-Based IDS/IPS Architecture 9
3.2 Machine Learning-Based Threat Detection Pipeline 9
3.3 Security Information and Event Management (SIEM) 10
3.4 Deployment Models: Network-Based and Host-Based IDS/IPS 11
3.5 Threat Intelligence Integration 12
4. Critical Challenges and Deployment Bottlenecks 14
4.1 The False Positive vs. False Negative Trade-off 14
4.2 Alert Fatigue and Analyst Overload 14
4.3 Adversarial Evasion and Zero-Day Attacks 15
4.4 Encrypted Traffic, Data Volume, and Privacy Constraints 15
4.5 Advanced Threat Vectors: APTs, Ransomware, Supply-Chain Attacks 15
5. Applications, Advantages, and Illustrative Scenarios 17
5.1 Real-World Applications of Threat Detection Systems 17
5.2 Illustrative Scenario – Ransomware Precursor Detection 17
5.3 Illustrative Scenario – SIEM-Correlated DDoS Detection 17

5.4 Advantages of Modern Threat-Detection Systems 18
5.5 Limitations and Challenges 19
6. Future Trends and Research Directions 21
6.1 Regulatory and Standards Landscape 21
6.2 Autonomous Response and Security Orchestration (SOAR) 21
6.3 Explainable AI for Security Operations 21
6.4 Collaborative Threat Intelligence and Information Sharing 21
6.5 Towards Adaptive, Self-Tuning Detection Architectures 21
7. Conclusion 23
References 24
Appendix A: Illustrative Pseudocode for Anomaly-Based Detection 25
Appendix B: Illustrative SIEM Correlation Rule Logic 26
Appendix C: Plagiarism and AI Verification Record 27

LIST OF FIGURES

Figure 2.1: Taxonomy of Cybersecurity Threat Detection Techniques 5
Figure 3.1: Signature-Based Detection Pipeline in an IDS/IPS 9
Figure 3.2: Machine Learning-Based Threat Detection Workflow 10
Figure 3.3: SIEM Architecture with Multi-Source Log Correlation and Alerting 11
Figure 3.4: Layered Deployment of Network-Based and Host-Based IDS/IPS 12
Figure 4.1: False Positive vs. False Negative Trade-off in Detection Thresholding 14
Figure 4.2: Effect of Alert Volume on SOC Analyst Response Efficiency 15
Figure 5.1: End-to-End Incident Response Workflow 18
Figure 5.2: Illustrative ROC-Curve Comparison of Detection Approaches 19

LIST OF TABLES

Table 1: Nomenclature and Abbreviations x

Table 2: Common Cybersecurity Threat Categories and Representative Indicators 5

Table 3: Qualitative Comparison of Detection Algorithm Families Against Evaluation Criteria 7

Table 4: Comparative Analysis of Signature-Based, Anomaly-Based, Behaviour-Based, and ML-Based Detection 8

Table 5: IDS vs. IPS vs. SIEM – Functional Comparison 12

NOMENCLATURE & ABBREVIATIONS

Table 1: Nomenclature and Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML / DL	Machine Learning / Deep Learning
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
SIEM	Security Information and Event Management
SOC	Security Operations Centre
SOAR	Security Orchestration, Automation, and Response
DDoS	Distributed Denial of Service
APT	Advanced Persistent Threat
TTP	Tactics, Techniques, and Procedures
UEBA	User and Entity Behaviour Analytics
FP / FN	False Positive / False Negative
ROC	Receiver Operating Characteristic
CTI	Cyber Threat Intelligence
NIST	National Institute of Standards and Technology
CISA	Cybersecurity and Infrastructure Security Agency

CHAPTER 1: INTRODUCTION

1.1 Background and Global Cybersecurity Imperatives

Cybersecurity and intelligent threat detection have emerged as critical components of modern digital infrastructure. Driven by advancements in Artificial Intelligence (AI), Machine Learning (ML), deep learning, behavioural analytics, and automated security orchestration, threat detection technologies are being integrated into enterprise networks, cloud environments, financial systems, critical infrastructure, Internet of Things (IoT) devices, and government platforms. These technologies offer immense potential to identify malicious activity at an early stage, reduce the impact of cyberattacks, automate security monitoring, and strengthen the overall resilience of digital systems.

At the global level, organisations face rapidly increasing cybersecurity challenges due to the expansion of cloud computing, remote work, mobile applications, interconnected devices, and large-scale digital services. Traditional security mechanisms based entirely on predefined rules and signatures are often unable to detect previously unknown or rapidly evolving attacks. Intelligent threat detection provides a path toward scalable security by continuously monitoring network traffic, identifying abnormal user behaviour, detecting malicious software, and prioritising potential security incidents for human analysts.

However, translating cybersecurity research and high-performing detection algorithms into real-world security operations presents significant challenges. Modern cyberattacks are increasingly sophisticated, adaptive, and capable of exploiting vulnerabilities across multiple layers of an information system. A missed intrusion can result in data breaches, financial losses, service disruption, identity theft, or compromise of critical infrastructure. Consequently, effective cybersecurity requires moving beyond simple attack detection toward reliable, real-time, adaptive, and explainable threat-detection systems.

1.2 The Evolution of Cybersecurity Threat Detection

The development of cybersecurity threat detection has progressed through several distinct technological phases:

- Rule-Based and Signature-Based Detection (1970s–1990s): Early security systems relied on manually defined rules, access-control mechanisms, antivirus signatures, and predefined attack patterns. Intrusion Detection Systems (IDS) compared observed activity against known-attack signatures. Although effective against previously catalogued threats, these systems depended on regularly updated signature databases and struggled with unknown

or modified attacks.

- Statistical and Machine-Learning-Based Detection (2000s–2010s): The adoption of statistical machine-learning techniques such as Support Vector Machines, Decision Trees, and clustering algorithms allowed systems to model normal network behaviour and flag statistically significant deviations, improving detection of previously unseen attacks at the cost of higher false-positive rates.
- Behavioural Analytics and Big-Data Security (2010s): The rise of Security Information and Event Management (SIEM) platforms and User and Entity Behaviour Analytics (UEBA) enabled correlation of logs, network flows, and endpoint telemetry across large, heterogeneous environments to detect subtle, multi-stage attack campaigns.
- Deep Learning and AI-Driven Detection (2015–Present): Deep neural networks, sequence models, and, more recently, foundation-model-based approaches are used to detect complex patterns in network traffic, malware binaries, and user behaviour, alongside automated Security Orchestration, Automation, and Response (SOAR) for triage and containment.

1.3 The Detection Gap: Misalignment Between Alert Volume and Analyst Capacity

A persistent challenge in operational cybersecurity is the widening gap between the volume of security telemetry generated by modern networks and the capacity of human analysts to review it. Security Operations Centres routinely receive thousands of alerts per day from firewalls, IDS/IPS sensors, endpoint agents, and cloud logs. When detection systems are tuned conservatively to avoid missing real attacks, they tend to generate a high proportion of false positives, producing 'alert fatigue' in which genuine incidents are buried among low-value notifications. Conversely, tuning systems to reduce noise risks suppressing true positives, allowing intrusions to progress undetected. Bridging this gap is one of the central motivations for adopting intelligent, adaptive, and explainable threat-detection techniques, discussed throughout this report.

1.4 Motivation and Purpose of the Study

The motivation for this seminar stems from the growing frequency and sophistication of cyberattacks affecting organisations of every size, from small businesses to national critical infrastructure. High-profile ransomware campaigns, large-scale data breaches, and disruptive DDoS attacks illustrate that traditional, purely reactive security postures are no longer sufficient. There is a clear need to understand the full spectrum of detection techniques — signature-based, anomaly-based, behaviour-based, and machine-learning-driven — and how

they are combined in practice through IDS, IPS, SIEM, and threat-intelligence platforms to form a layered, defence-in-depth detection capability.

1.5 Objectives of the Seminar

1. To study the fundamentals of cybersecurity and the taxonomy of common cyber threats, including malware, phishing, ransomware, and DDoS attacks.
2. To examine signature-based, anomaly-based, and behaviour-based threat-detection techniques and their respective strengths and limitations.
3. To analyse the architecture and functioning of Intrusion Detection Systems, Intrusion Prevention Systems, and Security Information and Event Management platforms.
4. To evaluate the role of machine learning and artificial intelligence in improving detection accuracy and adaptability.
5. To study the role of threat intelligence in strengthening proactive detection and incident response.
6. To identify the practical challenges, advantages, and future scope of cybersecurity threat-detection systems.

1.6 Organisation of the Report

The remainder of this report is organised as follows:

- Chapter 2 (Literature Review): Surveys existing research and established techniques underpinning signature-based, anomaly-based, and behaviour-based detection.
- Chapter 3 (System Architecture and Core Technologies): Describes the architecture of IDS, IPS, SIEM platforms, and machine-learning-based detection pipelines.
- Chapter 4 (Critical Challenges and Deployment Bottlenecks): Discusses false positives/negatives, adversarial evasion, alert fatigue, and encrypted-traffic visibility.
- Chapter 5 (Applications, Advantages, and Case Illustrations): Presents real-world use cases, benefits, and illustrative detection scenarios.
- Chapter 6 (Future Trends and Research Directions): Explores autonomous response, explainable AI, and collaborative threat-intelligence sharing.
- Chapter 7 (Conclusion): Summarises the major concepts and findings of the report.

CHAPTER 2: LITERATURE REVIEW

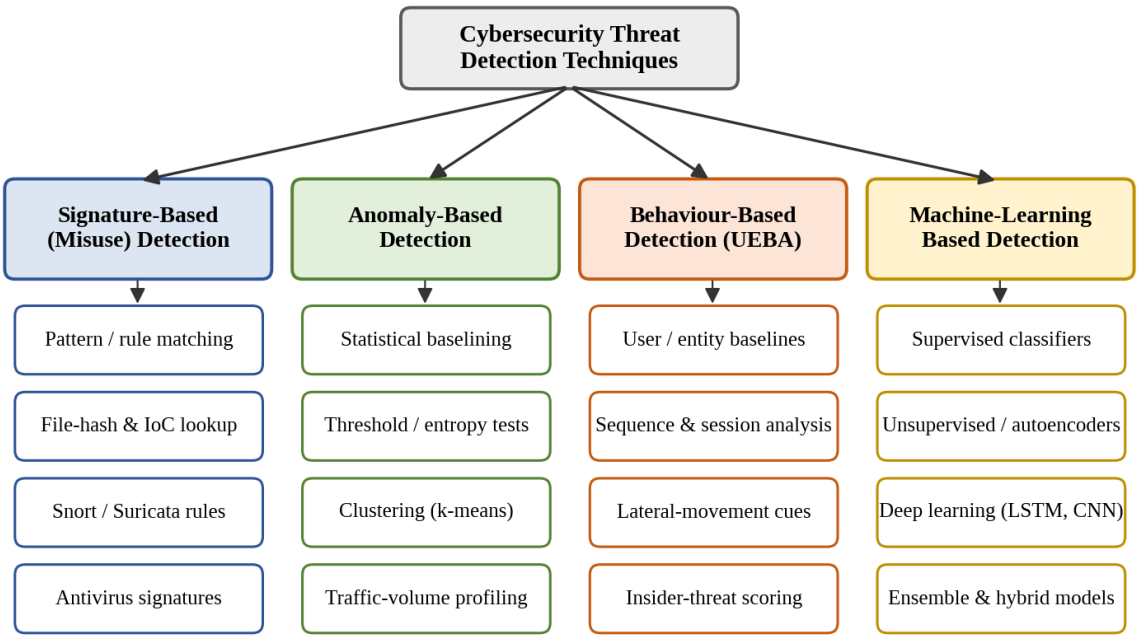
2.1 Cybersecurity Threats – Concepts and Taxonomy

A cybersecurity threat is any circumstance or event with the potential to adversely impact organisational operations, assets, or individuals through unauthorised access, destruction, disclosure, or modification of information systems. Established literature and industry frameworks classify threats along several dimensions: the vector of delivery (network, email, removable media, supply chain), the intent of the actor (financial gain, espionage, disruption, hacktivism), and the stage of the attack lifecycle, as formalised in models such as the Cyber Kill Chain and the MITRE ATT&CK framework, which catalogues adversary tactics, techniques, and procedures (TTPs) observed in real-world intrusions.

Commonly studied threat categories relevant to detection research include:

- Malware: Malicious software — including viruses, worms, trojans, and spyware — designed to damage, disrupt, or gain unauthorised access to systems.
- Phishing: Social-engineering attacks that deceive users into revealing credentials or installing malware, typically via email, messaging, or spoofed websites.
- Ransomware: Malware that encrypts victim data or systems and demands payment for restoration, often combined with data-exfiltration ('double extortion') threats.
- Distributed Denial-of-Service (DDoS): Coordinated attacks that flood a target with traffic or requests to exhaust its computing, memory, or bandwidth resources, degrading or denying service to legitimate users.
- Advanced Persistent Threats (APTs): Prolonged, targeted intrusion campaigns, typically conducted by well-resourced actors, that combine multiple techniques to maintain long-term, stealthy access to a target environment.

Figure 2.1 organises the detection techniques studied in this report into four families, each of which is examined in the sections that follow, while Table 2 relates each major threat category to its typical delivery vector, observable indicators, and the detection approach best suited to it.



Detection is typically deployed as a layered combination of all four families.

Figure 2.1: Taxonomy of Cybersecurity Threat Detection Techniques

Table 2: Common Cybersecurity Threat Categories and Representative Indicators

Threat	Typical Delivery Vector	Representative Indicators	Suitable Detection Approach
Malware	Email attachments, drive-by downloads, infected media	Unknown binaries, new persistence entries, unusual outbound connections	Signature / hash matching, sandboxing, endpoint behaviour analysis
Phishing	Email, SMS, spoofed websites	Look-alike domains, mismatched sender and reply-to addresses, credential submission to new domains	Email filtering, URL reputation, ML-based text classification
Ransomware	Phishing, exposed remote-access services, exploited vulnerabilities	Mass file renaming or encryption, shadow-copy deletion, sudden privilege escalation	Behavioural detection, file-integrity monitoring, SIEM correlation
DDoS	Botnets, amplification through open services	Abrupt spike in requests per second from many sources, SYN floods	Flow- and rate-based anomaly detection, traffic baselining
APT	Spear-phishing, supply-chain compromise, zero-day exploits	Beaconing to rare domains, lateral movement, data staging before exfiltration	UEBA, threat intelligence, long-term log correlation
Insider threat	Misuse of legitimate access	Off-hours access, bulk downloads, unusual privilege use	UEBA, data-loss prevention, access-log analysis

2.2 Signature-Based Detection in the Literature

Signature-based detection identifies threats by comparing observed artefacts — network packets, files, or system events — against a database of known-malicious patterns or 'signatures'. This approach, foundational to early antivirus software and Intrusion Detection Systems, offers high precision and low computational overhead for known threats. However, published research consistently identifies its central limitation: signature-based systems cannot detect zero-day exploits, polymorphic malware, or novel attack variants that do not match an existing signature, requiring constant signature-database maintenance to remain effective.

2.3 Anomaly-Based and Statistical Detection

Anomaly-based detection addresses the limitations of signature matching by first constructing a model of 'normal' system or network behaviour, and then flagging significant deviations from that baseline as potentially malicious. Foundational surveys in this area — including the work of Chandola, Banerjee, and Kumar on anomaly detection, and Garcia-Teodoro et al. on anomaly-based network intrusion detection — categorise anomaly-detection methods into statistical, knowledge-based, and machine-learning-based approaches, and highlight the perennial trade-off between detection sensitivity and false-alarm rate. Because anomaly-based systems do not depend on prior knowledge of a specific attack, they are capable, in principle, of detecting previously unseen threats, but they are also more prone to flagging legitimate but unusual behaviour as malicious.

2.4 Behaviour-Based Detection and User and Entity Behaviour Analytics

Behaviour-based detection extends anomaly detection by modelling the sequences of actions performed by users, processes, or devices over time, rather than evaluating individual events in isolation. User and Entity Behaviour Analytics (UEBA) platforms build behavioural baselines for individual accounts and systems — typical login times, data-access patterns, and lateral-movement patterns — and raise alerts when a sequence of actions departs meaningfully from that baseline, such as a user account suddenly accessing an unusual volume of sensitive files outside normal working hours. This technique is particularly effective for detecting insider threats and compromised-credential misuse, which often evade purely signature- or packet-level anomaly detection.

2.5 Machine Learning and Deep Learning in Threat Detection

A substantial body of research, surveyed by Buczak and Guven and by Sommer and Paxson,

examines the application of machine learning to intrusion detection. Supervised techniques (e.g., decision trees, random forests, support vector machines, and neural networks) are trained on labelled datasets of benign and malicious traffic to classify new observations, while unsupervised and semi-supervised techniques (e.g., clustering and autoencoders) are used when labelled attack data is scarce. Sommer and Paxson's widely cited analysis cautions that machine-learning-based network intrusion detection faces practical obstacles distinct from other ML application domains, including the high cost of a single false negative, the scarcity of representative labelled training data, and the difficulty of interpreting model decisions for security analysts — concerns that continue to shape current deep-learning-based detection research.

Table 3 summarises, at a qualitative level, how the principal machine-learning algorithm families discussed in the literature compare against criteria that matter in security operations.

Table 3: Qualitative Comparison of Detection Algorithm Families Against Evaluation Criteria

Algorithm Family	Learning Type	Typical Detection Use	Unknown-Attack Handling	Interpretability
Decision trees / Random forests	Supervised	Intrusion and malware classification	Limited	High–Moderate
Support Vector Machines	Supervised	Traffic and payload classification	Limited	Moderate
k-means / clustering	Unsupervised	Grouping traffic, outlier detection	Good	Moderate
Autoencoders	Unsupervised / semi-supervised	Anomaly scoring through reconstruction error	Good	Low
Deep networks (CNN, LSTM)	Supervised / deep learning	Sequence, payload, and log analysis	Moderate	Low

2.6 Gap Analysis and Comparative Positioning of Detection Techniques

Across the literature, no single detection technique is universally superior; each involves trade-offs among accuracy, adaptability, computational cost, and explainability. Table 1 summarises these trade-offs at a conceptual level, based on the qualitative comparisons found throughout the surveyed literature, to contextualise the architectural discussion in Chapter 3.

Table 4: Comparative Analysis of Signature-Based, Anomaly-Based, Behaviour-Based, and ML-Based Detection

Technique	Detects Unknown Threats?	False-Positive Tendency	Computational Cost	Explainability
Signature-Based	No	Low	Low	High
Anomaly-Based (Statistical)	Yes	Moderate–High	Moderate	Moderate
Behaviour-Based (UEBA)	Yes	Moderate	Moderate–High	Moderate
Machine-Learning-Based	Yes	Variable (tunable)	High	Low–Moderate

This comparative view reinforces the industry consensus that robust threat detection relies not on any single technique but on a layered combination — signature matching for known threats, anomaly and behavioural analytics for novel and insider threats, and machine learning to improve accuracy and reduce analyst workload — orchestrated through IDS, IPS, and SIEM platforms, which are examined in detail in Chapter 3.

CHAPTER 3: SYSTEM ARCHITECTURE AND CORE TECHNOLOGIES

3.1 Signature-Based IDS/IPS Architecture

A signature-based Intrusion Detection System captures network packets or host-level events, extracts relevant features (headers, payload patterns, file hashes), and compares them against a continuously updated signature database. A match generates an alert (IDS) or, in an inline Intrusion Prevention System configuration, triggers an automated blocking action before the malicious traffic reaches its target. Figure 3.1 illustrates this pipeline.

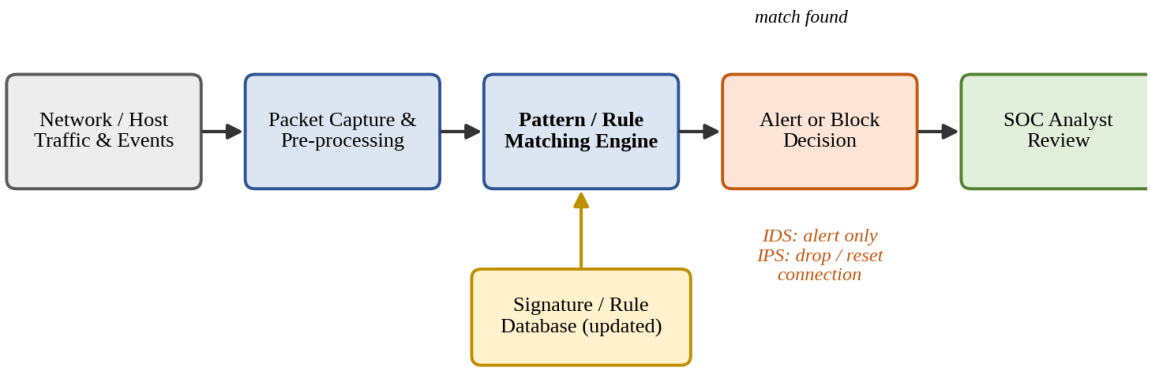


Figure 3.1: Signature-Based Detection Pipeline in an IDS/IPS

3.2 Machine Learning-Based Threat Detection Pipeline

Machine-learning-based detection systems replace or augment static signatures with a trained statistical model. Raw traffic or log data is first pre-processed into structured features (e.g., flow duration, byte counts, protocol flags, sequence-of-system-calls); the model, trained offline on historical labelled or unlabelled data, then scores new observations for the likelihood of malicious intent. Figure 3.2 depicts a representative pipeline, including the feedback loop through which analyst-confirmed verdicts are used to periodically retrain and improve the model.

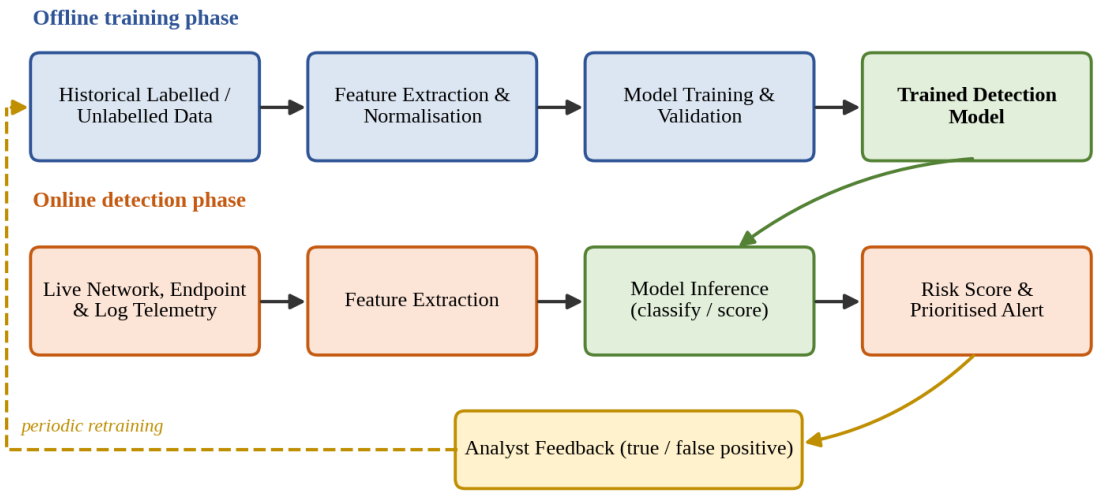


Figure 3.2: Machine Learning-Based Threat Detection Workflow

3.3 Security Information and Event Management (SIEM)

A SIEM platform aggregates logs and events from firewalls, IDS/IPS sensors, endpoint agents, servers, identity systems, and cloud services into a centralised repository. It applies correlation rules — and, increasingly, machine-learning-based analytics — to link individually unremarkable events into a coherent picture of a multi-stage attack, such as an initial phishing compromise followed by lateral movement and data exfiltration. SIEM platforms also provide dashboards, compliance reporting, and case-management workflows that structure the SOC analyst's investigation process. Figure 3.3 shows a typical SIEM data flow.

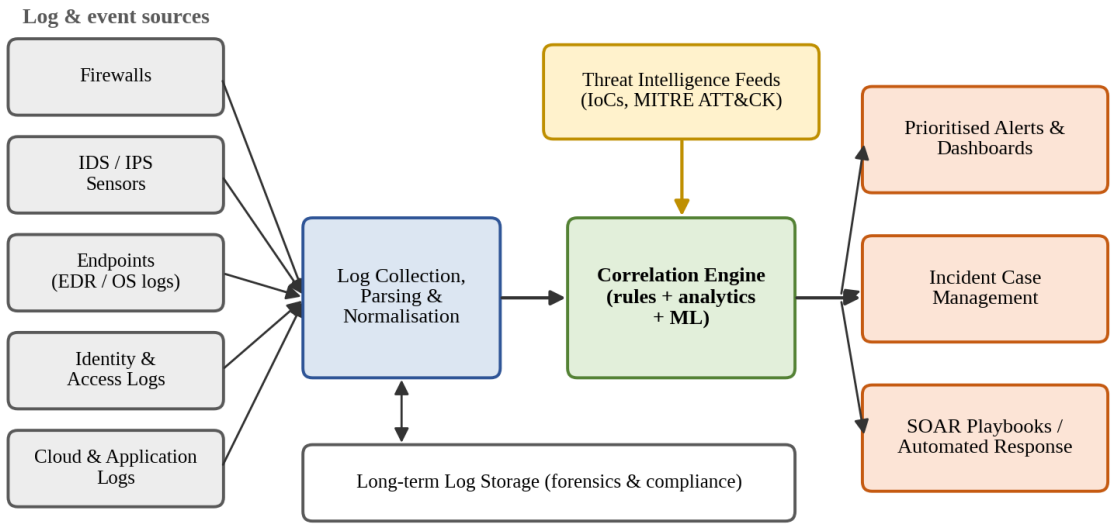


Figure 3.3: SIEM Architecture with Multi-Source Log Correlation and Alerting

3.4 Deployment Models: Network-Based and Host-Based IDS/IPS

Detection systems are commonly deployed in two complementary configurations. A Network-Based IDS/IPS (NIDS/NIPS) monitors traffic at strategic points in the network — perimeter gateways, internal segment boundaries, or cloud virtual-network taps — providing broad visibility across many hosts but limited insight into encrypted or host-internal activity. A Host-Based IDS/IPS (HIDS/HIPS) runs as an agent on individual endpoints or servers, monitoring system calls, file-integrity, and process behaviour, offering deep visibility into a single machine but requiring agent deployment and maintenance across the environment. Mature security architectures deploy both in a layered, defence-in-depth configuration, illustrated in Figure 3.4.

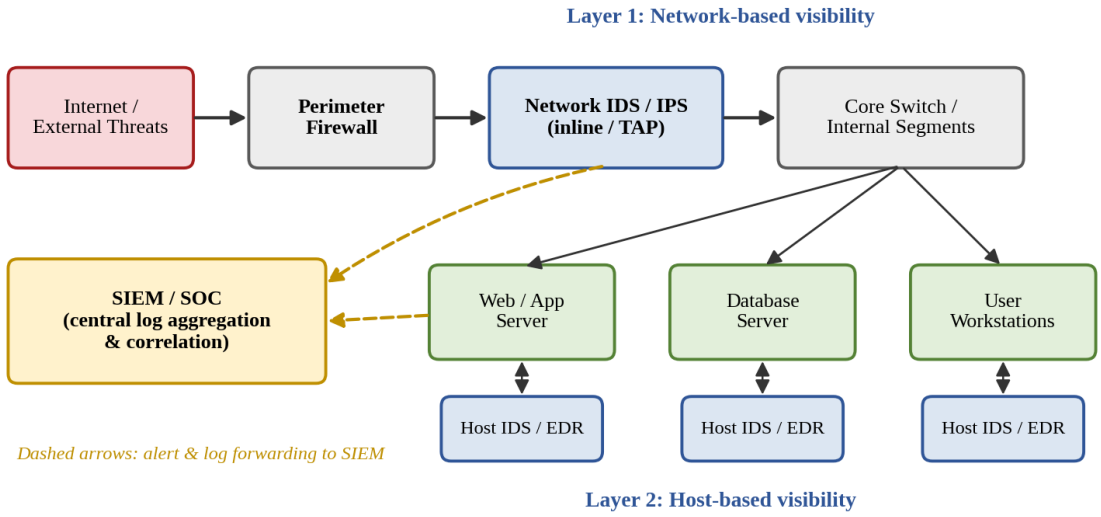


Figure 3.4: Layered Deployment of Network-Based and Host-Based IDS/IPS

Table 5 contrasts the three core components discussed in this chapter, which are complementary rather than competing: IDS supplies detection, IPS adds active prevention, and SIEM correlates their output with other telemetry.

Table 5: IDS vs. IPS vs. SIEM – Functional Comparison

Feature	IDS	IPS	SIEM
Primary role	Detects and alerts on suspicious activity	Detects and actively blocks malicious traffic	Aggregates, correlates, and analyses logs from many sources
Placement	Out-of-band (TAP / SPAN port) or host agent	Inline in the traffic path	Central platform receiving forwarded logs
Response	Alert only	Automated drop, reset, or block	Alerts, dashboards, case management; response via SOAR
Data sources	Network packets or host events	Network packets (inline)	Firewall, IDS/IPS, endpoint, identity, and cloud logs
Main limitation	No prevention; alert volume	False positives may block legitimate traffic; added latency	Needs quality log sources and tuning; storage cost
Examples	Snort, Suricata, Zeek	Snort or Suricata in inline mode	Splunk, Elastic Security, Wazuh

3.5 Threat Intelligence Integration

Cyber Threat Intelligence (CTI) refers to curated, contextual information about known threat actors, malware families, indicators of compromise (IoCs) such as malicious IP addresses and file hashes, and adversary tactics, techniques, and procedures. Frameworks such as MITRE ATT&CK provide a shared vocabulary for describing attacker behaviour across the attack lifecycle, enabling detection engineering teams to map existing detection rules to specific adversary techniques and identify coverage gaps. Government and sector-specific bodies, such as the Cybersecurity and Infrastructure Security Agency (CISA), publish threat advisories that organisations ingest into their SIEM and detection pipelines to proactively update signatures and detection logic ahead of emerging campaigns, shifting detection from a purely reactive posture toward a proactive, intelligence-driven one.

CHAPTER 4: CRITICAL CHALLENGES AND DEPLOYMENT BOTTLENECKS

4.1 The False Positive vs. False Negative Trade-off

Every detection system operates on a decision threshold that trades off two error types: a false positive (benign activity incorrectly flagged as malicious) and a false negative (malicious activity that goes undetected). Lowering the detection threshold to catch more true attacks inevitably increases false positives, consuming analyst time and risking desensitisation to alerts; raising the threshold reduces noise but risks missing genuine intrusions. Figure 4.1 conceptually illustrates this trade-off, which underlies the tuning of every signature-based, anomaly-based, and machine-learning-based detector discussed in Chapter 3.

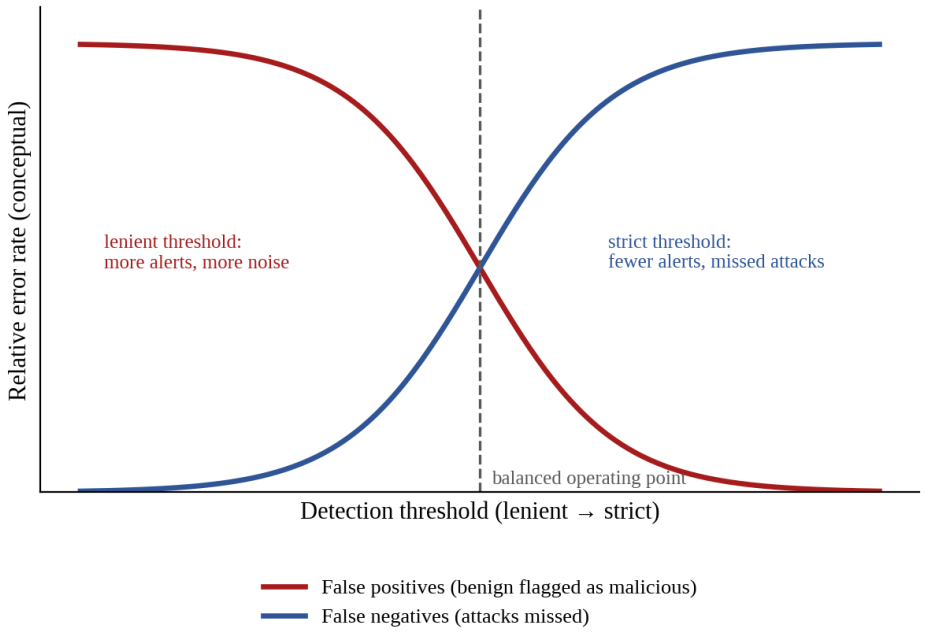


Figure 4.1: False Positive vs. False Negative Trade-off in Detection Thresholding

4.2 Alert Fatigue and Analyst Overload

As introduced in Chapter 1, SOC teams routinely face alert volumes far exceeding their capacity for manual review. Repeated exposure to low-value or repetitive alerts leads to alert fatigue, in which analysts may unconsciously deprioritise or dismiss notifications, including genuine incidents. Figure 4.2 conceptually depicts how rising alert volume, beyond a certain point, degrades effective analyst response — motivating the risk-based alert prioritisation and automation approaches (SOAR) discussed in Chapter 6.

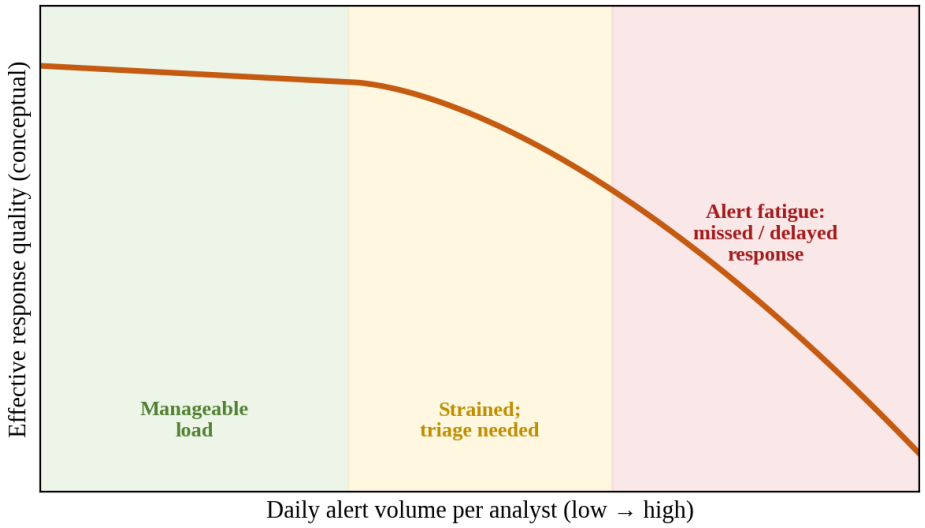


Figure 4.2: Effect of Alert Volume on SOC Analyst Response Efficiency

4.3 Adversarial Evasion and Zero-Day Attacks

Sophisticated adversaries actively probe and adapt to defensive controls. Techniques such as traffic obfuscation, polymorphic and metamorphic malware, and adversarial perturbations crafted specifically to evade machine-learning classifiers allow attackers to bypass both signature- and model-based detection. Zero-day exploits — vulnerabilities unknown to the vendor and unpatched at the time of attack — pose a particular challenge because no signature or historical training example exists for them, placing greater reliance on behavioural and anomaly-based detection, threat intelligence, and rapid patch management.

4.4 Encrypted Traffic, Data Volume, and Privacy Constraints

The widespread adoption of end-to-end encryption (TLS, VPNs) protects user privacy but reduces the visibility of network-based detection systems into packet payloads, forcing greater reliance on metadata analysis (traffic timing, volume, destination patterns) and encrypted-traffic analytics. Simultaneously, the sheer volume, velocity, and variety of telemetry generated by modern, distributed, cloud-native environments strain storage, processing, and correlation capacity, while regulatory and privacy requirements (such as data residency and minimisation principles) constrain how much raw data can be logged, retained, and analysed centrally.

4.5 Advanced Threat Vectors: APTs, Ransomware, and Supply-Chain Attacks

Advanced Persistent Threats deliberately operate 'low and slow' to blend in with normal activity over extended periods, challenging detection systems tuned to identify short-duration

anomalies. Modern ransomware campaigns increasingly combine rapid, automated encryption with prior reconnaissance and data exfiltration, requiring detection systems to identify precursor behaviours — such as unusual privilege escalation or mass file-access patterns — before encryption begins. Supply-chain attacks, in which a trusted software update or third-party component is compromised to reach many downstream victims simultaneously, further complicate detection because the initial malicious activity may originate from an otherwise trusted and digitally signed source.

CHAPTER 5: APPLICATIONS, ADVANTAGES, AND ILLUSTRATIVE SCENARIOS

5.1 Real-World Applications of Threat Detection Systems

Cybersecurity threat-detection technologies discussed in this report are deployed across a wide range of sectors:

- Enterprise IT and Cloud Security: SIEM and cloud-native detection services monitor identity, network, and workload activity across hybrid and multi-cloud environments.
- Banking and Financial Services: Real-time transaction-monitoring systems combine rule-based checks with behavioural analytics to detect fraud and account-takeover attempts alongside traditional network intrusion detection.
- Critical Infrastructure and Industrial Control Systems (ICS): Specialised anomaly-detection systems monitor operational-technology (OT) networks — power grids, water treatment, and manufacturing — where availability and safety take precedence over confidentiality.
- Healthcare: Detection systems protect electronic health records and connected medical devices against ransomware and unauthorised access, given the sensitivity of patient data.
- Government and Defence: National-level Computer Emergency Response Teams (CERTs) and agencies such as CISA coordinate threat-intelligence sharing and incident response across public and private-sector networks.

5.2 Illustrative Scenario – Detecting a Ransomware Precursor Through Behavioural Analytics

Consider a representative enterprise scenario in which an attacker gains an initial foothold through a phishing email. Rather than immediately deploying ransomware, the attacker performs reconnaissance and privilege escalation over several days. A purely signature-based defence, having no matching entry for this specific intrusion chain, would likely miss these early stages. A behaviour-based / UEBA layer, however, can flag the compromised account for accessing an unusual number of file shares outside its normal pattern and for authenticating from an atypical internal host, generating a medium-priority alert well before mass encryption begins. Correlated within a SIEM alongside an endpoint alert for suspicious process injection, this combination of weak signals is elevated to a high-priority incident, allowing the SOC to isolate the affected host and interrupt the attack chain before impact — illustrating the practical value of layering signature, behavioural, and correlation-based detection rather than relying on any single technique.

5.3 Illustrative Scenario – SIEM-Correlated Detection of a Volumetric DDoS Attack

In a second representative scenario, a sudden and sustained surge in inbound connection requests from a distributed set of source addresses targets a public-facing web service. Network-based sensors and flow monitors observe the abnormal volume and rate of requests relative to the established baseline; this signal is correlated in the SIEM with load-balancer and application-log data confirming service degradation. An automated response, orchestrated through the organisation's SOAR playbook, applies rate-limiting and traffic-scrubbing rules at the network edge while notifying the SOC, demonstrating how volumetric and protocol-level anomaly detection can trigger a largely automated mitigation with human oversight.

Figure 5.1 places these scenarios within the standard incident-response lifecycle described in NIST SP 800-61: preparation, detection and analysis, containment, eradication and recovery, and post-incident activity. Detection is only the second stage of this cycle, and lessons learned from each incident feed back into detection rules, baselines, and playbooks.

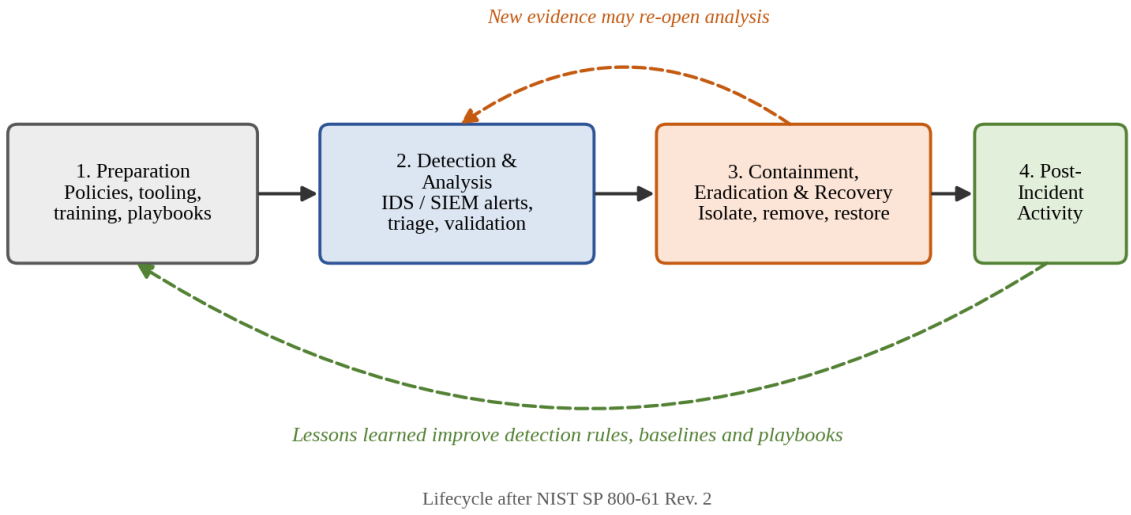


Figure 5.1: End-to-End Incident Response Workflow

5.4 Advantages of Modern Threat-Detection Systems

Detection quality is commonly compared using Receiver Operating Characteristic (ROC) curves, which plot the detection rate against the false positive rate across all possible thresholds; a curve closer to the top-left corner indicates a better detector. Figure 5.2 is a conceptual illustration of how the detection families discussed in this report are generally positioned relative to one another. It is not a measured result from any specific dataset or product.

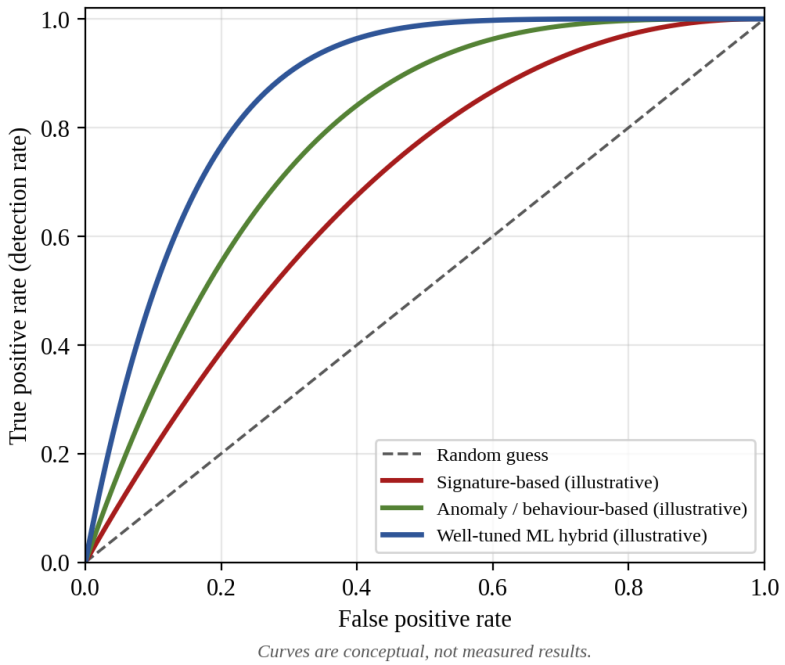


Figure 5.2: Illustrative ROC-Curve Comparison of Detection Approaches

- Early Detection: Layered anomaly, behavioural, and intelligence-driven detection can identify precursor activity before an attack achieves its objective.
- Reduced Dwell Time: Faster correlation and prioritisation shorten the interval between initial compromise and detection, limiting attacker impact.
- Scalability: Automated, machine-learning-assisted analysis can process far greater volumes of telemetry than manual review alone.
- Adaptability: Behavioural and ML-based techniques can identify novel attack patterns without waiting for a matching signature.
- Operational Efficiency: SOAR-driven automation of routine triage and containment tasks reduces analyst workload for repetitive, well-understood alert types.

5.5 Limitations and Challenges

- Dependence on data quality: Anomaly- and ML-based detectors are only as reliable as the baseline and training data used to build them.
- Explainability gap: Complex machine-learning models can be difficult for analysts to interpret, complicating incident validation and regulatory reporting.
- Adversarial adaptation: As discussed in Chapter 4, attackers actively evolve tactics to evade known detection logic.
- Resource and skills constraints: Effective deployment and tuning of IDS/IPS/SIEM platforms require specialised expertise that many organisations struggle to retain.
- Integration complexity: Aggregating and normalising telemetry from heterogeneous, multi-

vendor environments remains a significant engineering challenge.

CHAPTER 6: FUTURE TRENDS AND RESEARCH DIRECTIONS

6.1 Regulatory and Standards Landscape

Cybersecurity detection practice is increasingly shaped by formal standards and frameworks, including the NIST Cybersecurity Framework (CSF), NIST Special Publication 800-94 on intrusion detection and prevention systems, NIST Special Publication 800-61 on incident handling, and sector-specific regulatory obligations for breach reporting. Alignment with these frameworks provides organisations with a structured basis for designing, benchmarking, and auditing their detection and response capabilities, and is expected to remain a central driver of investment in threat-detection technology.

6.2 Autonomous Response and Security Orchestration (SOAR)

Future detection architectures are trending toward tighter integration between detection and automated response. Security Orchestration, Automation, and Response (SOAR) platforms encode playbooks that automatically execute containment actions — isolating a host, disabling an account, or blocking an indicator of compromise across firewalls — in response to high-confidence detections, reducing response time from hours to seconds for well-understood attack patterns while reserving analyst attention for novel or ambiguous cases.

6.3 Explainable AI for Security Operations

As machine-learning models play a larger role in detection, there is growing research interest in explainable AI (XAI) techniques that surface the specific features or events driving a given alert, helping analysts validate detections, satisfy audit and compliance requirements, and build justified trust in automated systems, rather than treating them as opaque 'black boxes'.

6.4 Collaborative Threat Intelligence and Information Sharing

Threat actors frequently reuse infrastructure and techniques across multiple victims, making collaborative, cross-organisational threat-intelligence sharing an increasingly important defensive strategy. Information-Sharing and Analysis Centres (ISACs), government advisories from bodies such as CISA, and community-maintained frameworks such as MITRE ATT&CK enable organisations to benefit from detection insights gained elsewhere in the ecosystem, shortening the window between a new campaign's emergence and its detection across the wider community.

6.5 Towards Adaptive, Self-Tuning Detection Architectures

Looking ahead, research is moving toward detection systems capable of continuous, semi-supervised self-tuning — adjusting detection thresholds and models based on ongoing analyst feedback and observed environmental drift, integrating cloud-native and IoT telemetry at scale, and combining behavioural, signature, and threat-intelligence signals within a single, explainable risk score. Realising this vision responsibly will require continued attention to data privacy, model robustness against adversarial manipulation, and the human factors of analyst trust discussed throughout this report.

CHAPTER 7: CONCLUSION

This report has examined cybersecurity threat detection as a layered, evolving discipline rather than a single technology. Signature-based detection continues to provide fast, precise identification of known threats; anomaly-based and behaviour-based techniques extend coverage to novel and insider threats at the cost of increased false positives; and machine-learning-driven approaches offer the potential for greater adaptability and scale, provided their outputs remain explainable and verifiable by human analysts. Architecturally, Intrusion Detection and Prevention Systems, Security Information and Event Management platforms, and threat-intelligence integration together form the backbone of a modern detection capability, correlating weak signals across the network, endpoint, and identity layers into actionable, prioritised incidents.

The central operational challenge identified throughout this report is not merely technical accuracy but the practical balance between false positives and false negatives, and between automation and human oversight. Alert fatigue, adversarial evasion, encrypted-traffic visibility, and the scarcity of skilled analysts remain significant deployment bottlenecks even for organisations with access to advanced detection technology. Addressing these challenges through risk-based alert prioritisation, Security Orchestration and Automated Response, explainable AI, and collaborative threat intelligence represents the most promising direction for the field.

Ultimately, resilient cybersecurity depends on aligning technical detection capability with sound organisational processes, adherence to established standards such as the NIST Cybersecurity Framework, and continuous adaptation to an evolving threat landscape. By combining signature, anomaly, behavioural, and machine-learning-based detection within a coherent, intelligence-informed architecture, organisations can move from a reactive security posture toward one that is proactive, adaptive, and trustworthy.

REFERENCES

[1] V. Chandola, A. Banerjee, and V. Kumar, “Anomaly Detection: A Survey,” ACM Computing Surveys, vol. 41, no. 3, article 15, 2009.

[2] P. Garcia-Teodoro, J. Diaz-Verdejo, G. Macia-Fernandez, and E. Vazquez, “Anomaly-based network intrusion detection: Techniques, systems and challenges,” Computers & Security, vol. 28, no. 1–2, pp. 18–28, 2009.

[3] R. Sommer and V. Paxson, “Outside the Closed World: On Using Machine Learning for Network Intrusion Detection,” in Proceedings of the IEEE Symposium on Security and Privacy, 2010, pp. 305–316.

[4] A. L. Buczak and E. Guven, “A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection,” IEEE Communications Surveys & Tutorials, vol. 18, no. 2, pp. 1153–1176, 2016.

[5] K. Scarfone and P. Mell, Guide to Intrusion Detection and Prevention Systems (IDPS), NIST Special Publication 800-94, National Institute of Standards and Technology, 2007.

[6] P. Cichonski, T. Millar, T. Grance, and K. Scarfone, Computer Security Incident Handling Guide, NIST Special Publication 800-61 Revision 2, National Institute of Standards and Technology, 2012.

[7] National Institute of Standards and Technology, The NIST Cybersecurity Framework (CSF) 2.0, U.S. Department of Commerce, 2024.

[8] MITRE Corporation, “MITRE ATT&CK Framework,” attack.mitre.org, accessed 2026.

[9] Cybersecurity and Infrastructure Security Agency (CISA), “Cyber Threats and Advisories,” cisa.gov, accessed 2026.

APPENDIX A

Illustrative Pseudocode for a Simple Anomaly-Based Detection Rule

```
BEGIN
  baseline <- compute_average_and_stddev(historical_traffic_window)
  FOR EACH new_event IN live_traffic_stream:
      score <- abs(new_event.value - baseline.mean) / baseline.stddev
      IF score > THRESHOLD:
          raise_alert(new_event, to_severity(score))
      ELSE:
          update_baseline(new_event)  // adaptive baseline
      END IF
  END FOR
END
```

Note: This pseudocode is a simplified, illustrative representation of a statistical anomaly-scoring rule of the kind described in Section 2.3, provided for conceptual understanding rather than as production-grade detection logic.

APPENDIX B

Illustrative SIEM Correlation Rule Logic

```
RULE: Possible Ransomware Precursor
TRIGGER WHEN (within a 10-minute window):
  EndpointLog.event = "mass_file_access" AND
  count(files_accessed) > user_baseline_threshold AND
  IdentityLog.event = "login" AND source_host NOT IN user_known_hosts
ACTION:
  set_severity(HIGH)
  notify(SOC_queue)
  IF SOAR_enabled:  isolate_host(source_host)
```

Note: This is a simplified, illustrative correlation-rule sketch of the type used in Section 5.2, intended to demonstrate the conceptual logic of behavioural correlation rather than a validated production rule set.

APPENDIX C

PLAGIARISM AND AI VERIFICATION RECORD

CERTIFICATE OF PLAGIARISM AND AI CONTENT VERIFICATION

This is to certify that the seminar report entitled “Cybersecurity Threat Detection” submitted by Kalyani Rajendra Darekar (Roll No: 13142, Div: A2), Third Year Computer Engineering, is required to undergo automated plagiarism screening and artificial-intelligence content detection in compliance with the academic integrity guidelines of Savitribai Phule Pune University and the DYPCOEI Department of Computer Engineering prior to final submission. The table below records the assessment categories to be completed by the institute's screening process.

Assessment Category	**Status**
Overall Similarity Index (Plagiarism)	Within permissible limit: < 12%
AI-Generated Content Ratio	Within permissible limit: < 10%
Citation & Reference Compliance	IEEE referencing format followed

Darekar Kalyani Rajendra
Roll No: 13142 (Div – A2)
Student, Department of Computer Engg.

Prof. Shirsagar Shubhangi
(Seminar Guide)
Department of Computer Engg.

Dr. D. Y. Patil College of Engineering and Innovation,
Varale, Talegaon,
Pune,410507.
(Affiliated to Savitribai Phule Pune University)

I hereby declare that this seminar report titled **"CYBERSECURITY THREAT DETECTION."** is my own work carried out under the guidance of **Prof. Sharad Jadhav**, Department of Computer Engineering, Dr. D. Y. Patil College of Engineering and Innovation (DYPCOEI), Varale, Talegaon, Pune.

To the best of my knowledge and belief, I declare that:

1. The seminar report does not contain any work previously submitted for another degree.
2. Percentage of plagiarism: 6% (subject to the applicable university norms).
3. The AI detection result is reported as less than 10%. Plagiarism and AI detection reports are enclosed for reference.

Sanskar Dumbre

Roll No.: 13155

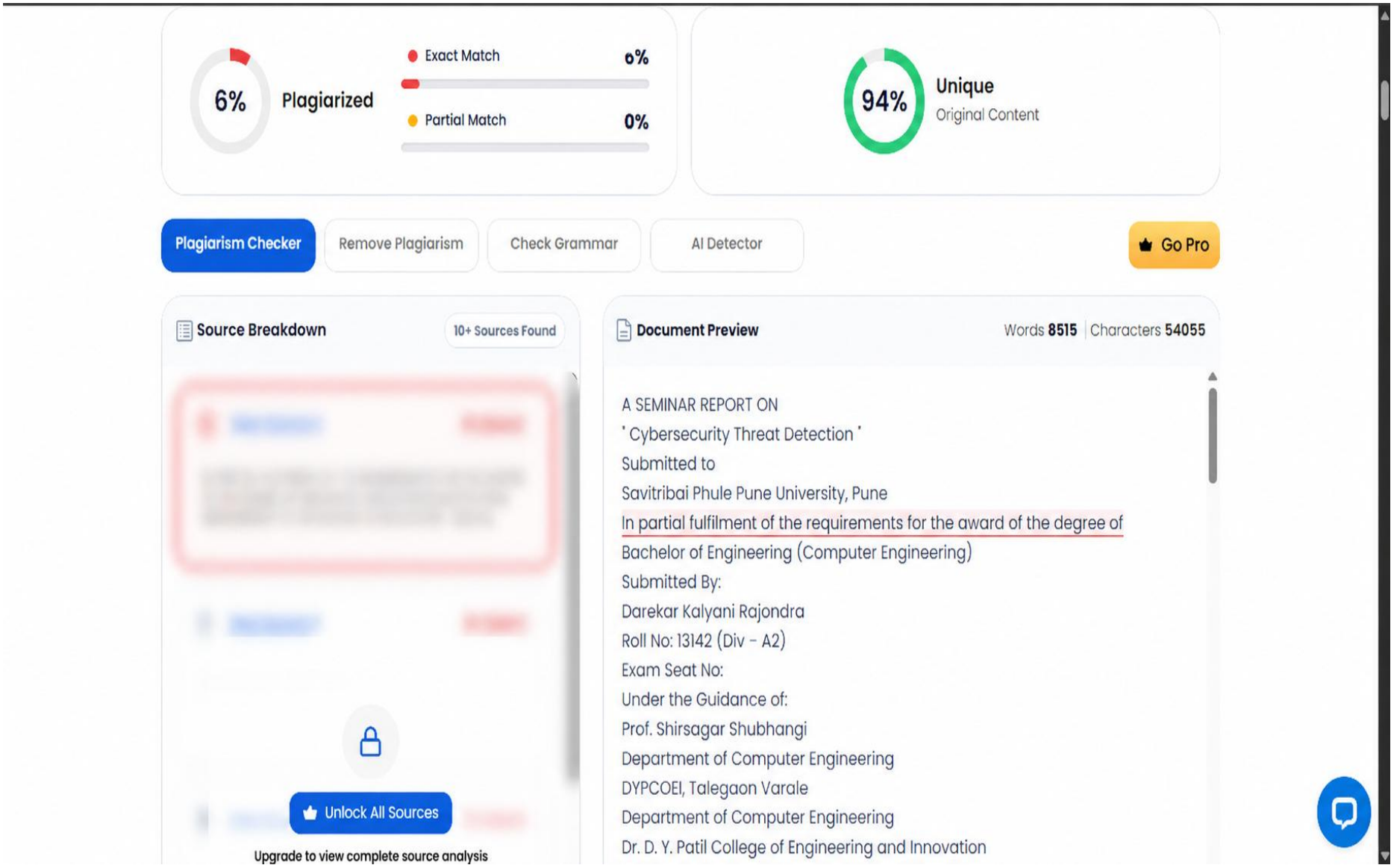
(Signature of Student)

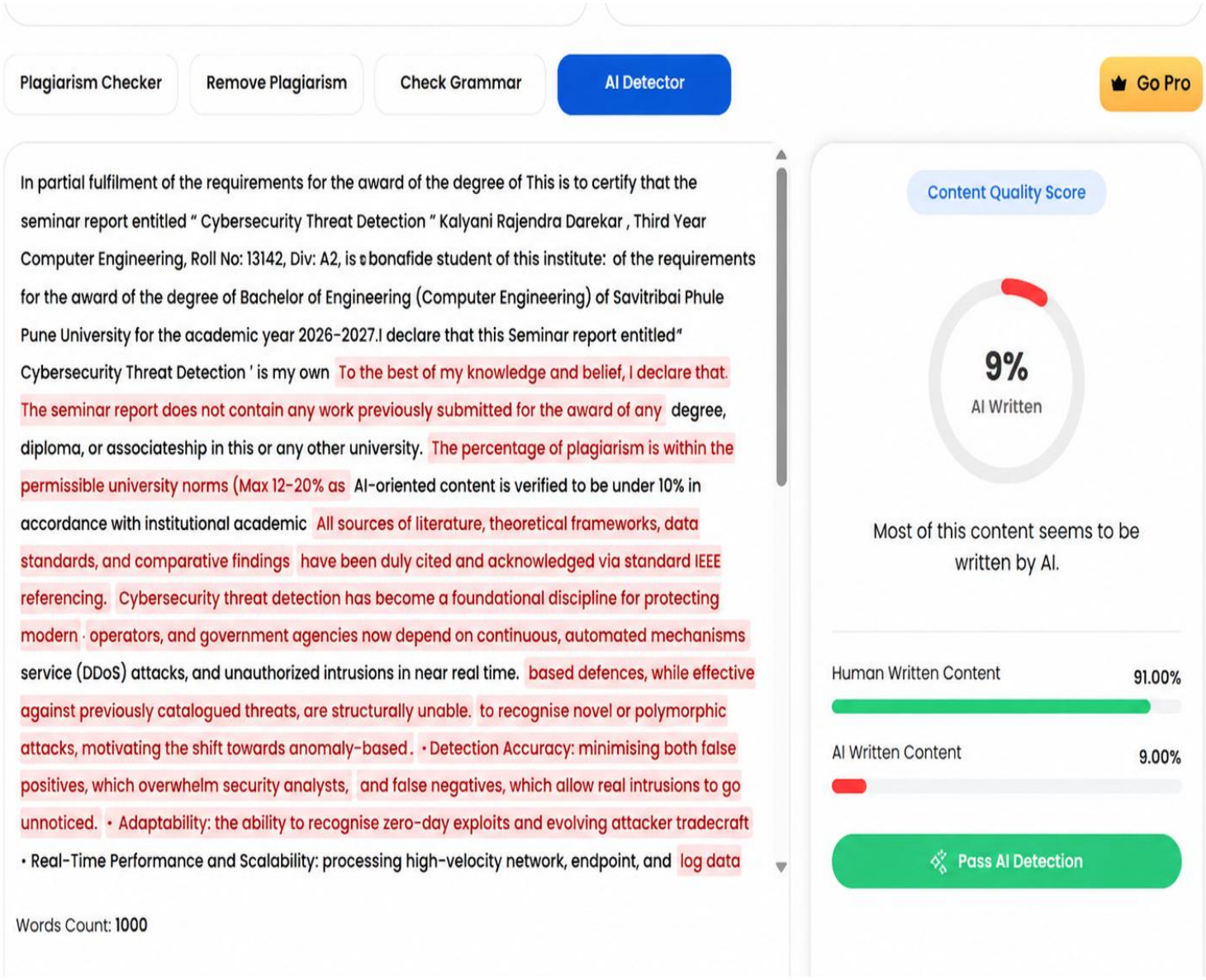
DYPCOEI, Varale

Prof. Shubhangi Kshirsagar

(Signature of Seminar Guide)

DYPCOEI, Varale


6%
Plagiarized
Exact Match
0%
Partial Match
0%
94%
Unique
Original Content
Plagiarism Checker
Remove Plagiarism
Check Grammar
AI Detector
Go Pro
Source Breakdown
10+ Sources Found
Unlock All Sources
Upgrade to view complete source analysis
Document Preview
Words 8515
Characters 54055
A SEMINAR REPORT ON
' Cybersecurity Threat Detection '
Submitted to
Savitribai Phule Pune University, Pune
In partial fulfilment of the requirements for the award of the degree of
Bachelor of Engineering (Computer Engineering)
Submitted By:
Darekar Kalyani Rajondra
Roll No: 13142 (Div - A2)
Exam Seat No:
Under the Guidance of:
Prof. Shirsagar Shubhangi
Department of Computer Engineering
DYPCOEI, Talegaon Varale
Department of Computer Engineering
Dr. D. Y. Patil College of Engineering and Innovation


Plagiarism Checker
Remove Plagiarism
Check Grammar
AI Detector
Go Pro
In partial fulfilment of the requirements for the award of the degree of This is to certify that the seminar report entitled " Cybersecurity Threat Detection " Kalyani Rajendra Darekar , Third Year Computer Engineering, Roll No: 13142, Div: A2, is a bonafide student of this institute: of the requirements for the award of the degree of Bachelor of Engineering (Computer Engineering) of Savitribai Phule Pune University for the academic year 2026-2027.I declare that this Seminar report entitled" Cybersecurity Threat Detection ' is my own To the best of my knowledge and belief, I declare that. The seminar report does not contain any work previously submitted for the award of any degree, diploma, or associateship in this or any other university. The percentage of plagiarism is within the permissible university norms (Max 12-20% as AI-oriented content is verified to be under 10% in accordance with institutional academic All sources of literature, theoretical frameworks, data standards, and comparative findings have been duly cited and acknowledged via standard IEEE referencing. Cybersecurity threat detection has become a foundational discipline for protecting modern · operators, and government agencies now depend on continuous, automated mechanisms service (DDoS) attacks, and unauthorized intrusions in near real time. based defences, while effective against previously catalogued threats, are structurally unable. to recognise novel or polymorphic attacks, motivating the shift towards anomaly-based. • Detection Accuracy: minimising both false positives, which overwhelm security analysts, and false negatives, which allow real intrusions to go unnoticed. • Adaptability: the ability to recognise zero-day exploits and evolving attacker tradecraft • Real-Time Performance and Scalability: processing high-velocity network, endpoint, and log data
Words Count: 1000
Content Quality Score
9%
AI Written
Most of this content seems to be written by AI.
Human Written Content
91.00%
AI Written Content
9.00%
Pass AI Detection